Technology Watch: Answer Key for the SWOT Matrix Series

EXERCISE 01: AgriTech & IoT – Case of the Start-up TechDZ

1.1 Classification of the Findings in the SWOT Matrix

SWOT Matrix Category	Associated Findings & Descriptions
Strengths (S) **[Internal Factors +]**	• Finding 3: Reliable, robust products with low energy consumption. • Finding 7: Partnerships and collaborations with several local pilot farms.
Weaknesses (W) **[Internal Factors -]**	• Finding 2: Shortage of IoT skills and specialized staff within the TechDZ team. • Finding 4: Limited marketing and sales budget.
Opportunities (O) **[External Factors +]**	• Finding 1: Rapid growth of the connected agriculture market in Algeria. • Finding 5: Government support and public incentives for AgriTech projects.
Threats (T) **[External Factors -]**	• Finding 6: Strong foreign competition (low-cost Chinese solutions and high-end European ones). • Finding 8: Recurring Internet connectivity problems in rural areas.

1.2 Cross-Referenced Operational Strategic Decisions (TOWS Matrix)

• **Strategy SO (Offensive):** Leverage the partnerships with the pilot farms (S7) and the proven reliability of the products (S3) to apply for public subsidies and government incentives (O5), in order to finance large-scale deployment on the rapidly growing Algerian market (O1).

• **Strategy ST (Defensive):** Take advantage of the low energy consumption and robustness of the sensors (S3) to develop an Edge Computing architecture (local processing). This ensures continuous operation even during a network outage (T8) and creates a clear differentiation from foreign competition (T6).

• **Strategy WO (Adjustment):** Draw on government support and incubation programs (O5) to obtain targeted funding for IoT training/recruitment (W2) and to subsidize communication/marketing actions (W4).

• **Strategy WT (Survival):** Concentrate the sales effort on the network of pilot farms through word of mouth and direct selling (W4), while simplifying the sensors' software architecture to reduce dependence on specialized IoT recruitment (W2) and work around rural connectivity outages (T8).

1.3 Priority Major Axes of Technology Watch

1. Watch on LPWAN networks and embedded computing (Edge Computing): Monitor developments in low-bandwidth, long-range wireless protocols (LoRaWAN, NB-IoT) and in local storage and processing techniques, in order to ensure uninterrupted operation of the sensors in isolated rural areas.

2. Competitive and technological watch on low-cost IoT components: Closely track hardware innovations and the pricing strategies of international players (notably Asian ones) in order to preserve the price/performance competitiveness of the TechDZ modules.

EXERCISE 02: e-Health, Big Data & Sovereignty – Case of the Company HealthDZ

2.1 Main Objective of the HealthDZ Project

The main objective of HealthDZ is to design and deploy a connected health digital platform based on medical IoT and big data analytics (Big Data / AI). This solution enables real-time remote medical monitoring, early detection of physiological anomalies and prevention of pathological risks, perfectly suited to the constraints and specificities of the Algerian healthcare system.

2.2 SWOT Diagnosis of the Company HealthDZ

Analysis Axis	HealthDZ Diagnostic Elements
Strengths (S)	• Solid technical expertise of the founding team (electronics, embedded systems, Big Data). • Field validation through successful tests in several pilot hospitals (Algiers, Blida, Constantine). • Ergonomic multilingual interface (Arabic, French, English) suited to practitioners.
Weaknesses (W)	• Limited funding (own funds/competitions) restricting communication, maintenance and certification. • Heavy dependence on imported electronic components and supply lead times. • Use of foreign Cloud services raising data sovereignty and security issues. • Lack of in-house expertise specialized in cybersecurity and in medical regulation/law.
Opportunities (O)	• Strong state support for the digitalization of healthcare and local innovation. • Access to funding and modernization programs from international organizations (WHO, EU, IsDB). • Rapid expansion of the post-COVID e-health market and high smartphone ownership among citizens.
Threats (T)	• Strong competition from multinationals with already certified products and a large reputation. • Strengthened regulatory framework on the confidentiality and protection of medical data. • Uneven and unreliable Internet coverage in rural regions. • Resistance to change among part of the medical staff.

2.3 Concrete Strategic Actions for HealthDZ

• **Action 1 - Sovereign Cloud Partnership & Regulatory Compliance:** Migrate the hosting of medical data to sovereign data centers on national territory (e.g. Algérie Télécom) and form a strategic alliance with, or hire, an expert consultant in medical law/cybersecurity to ensure full compliance with Algerian personal data law.

• **Action 2 - Applying for International Funding & Medical Certification:** Submit support applications to the WHO, the European Union and the Islamic Development Bank to fund the rigorous medical certification process for the sensors. In parallel, organize free training workshops to overcome the resistance to change among healthcare professionals.